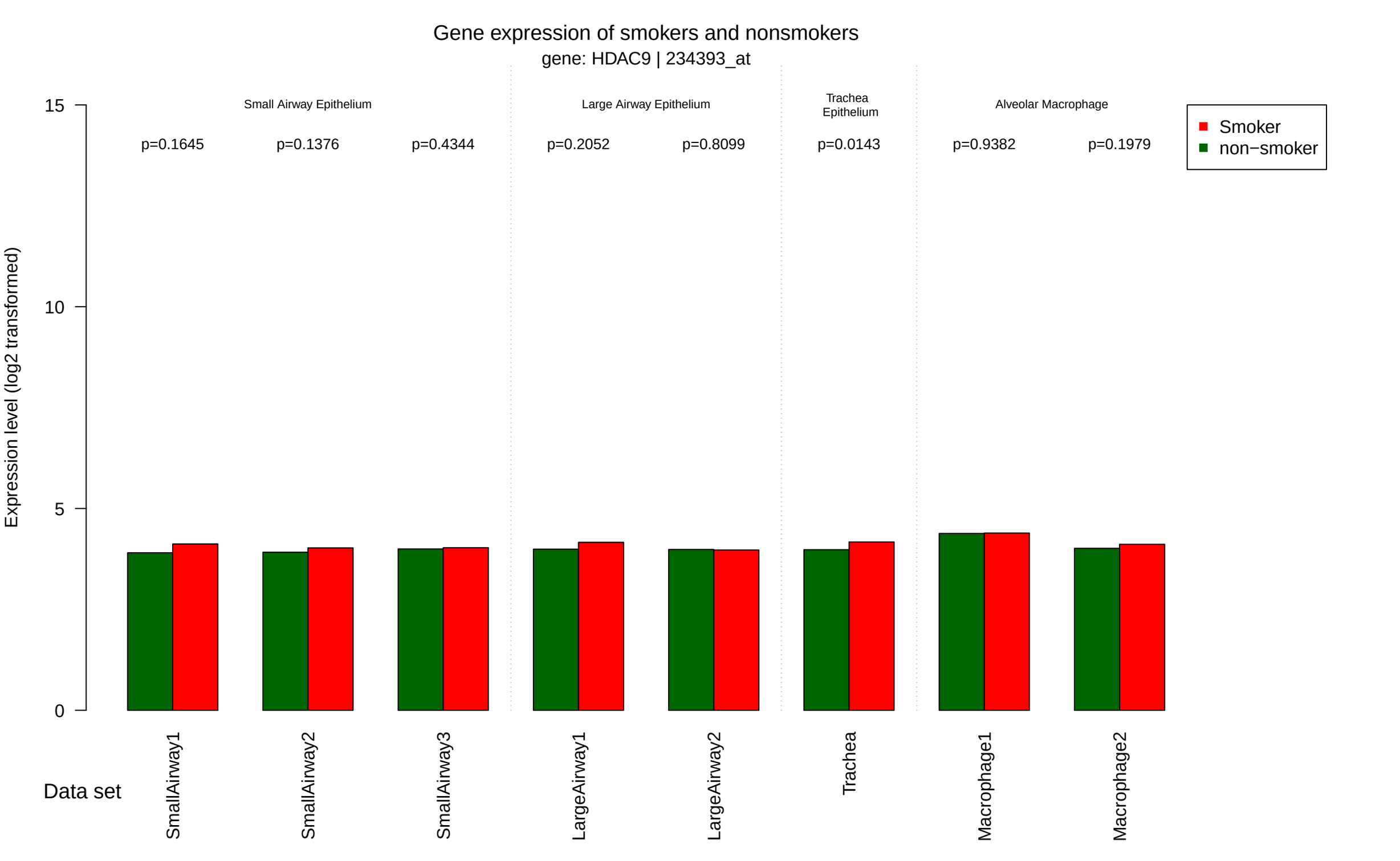
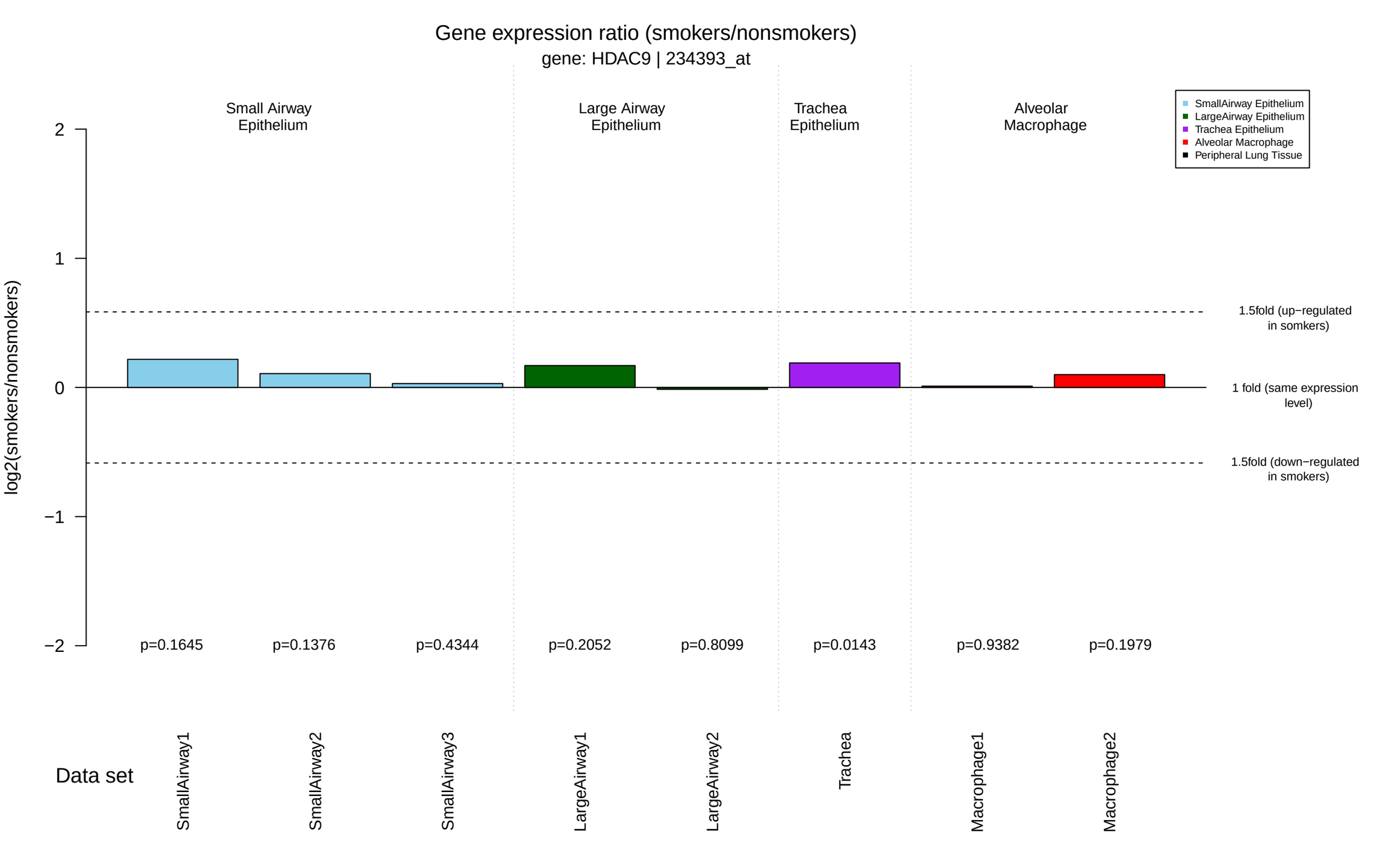
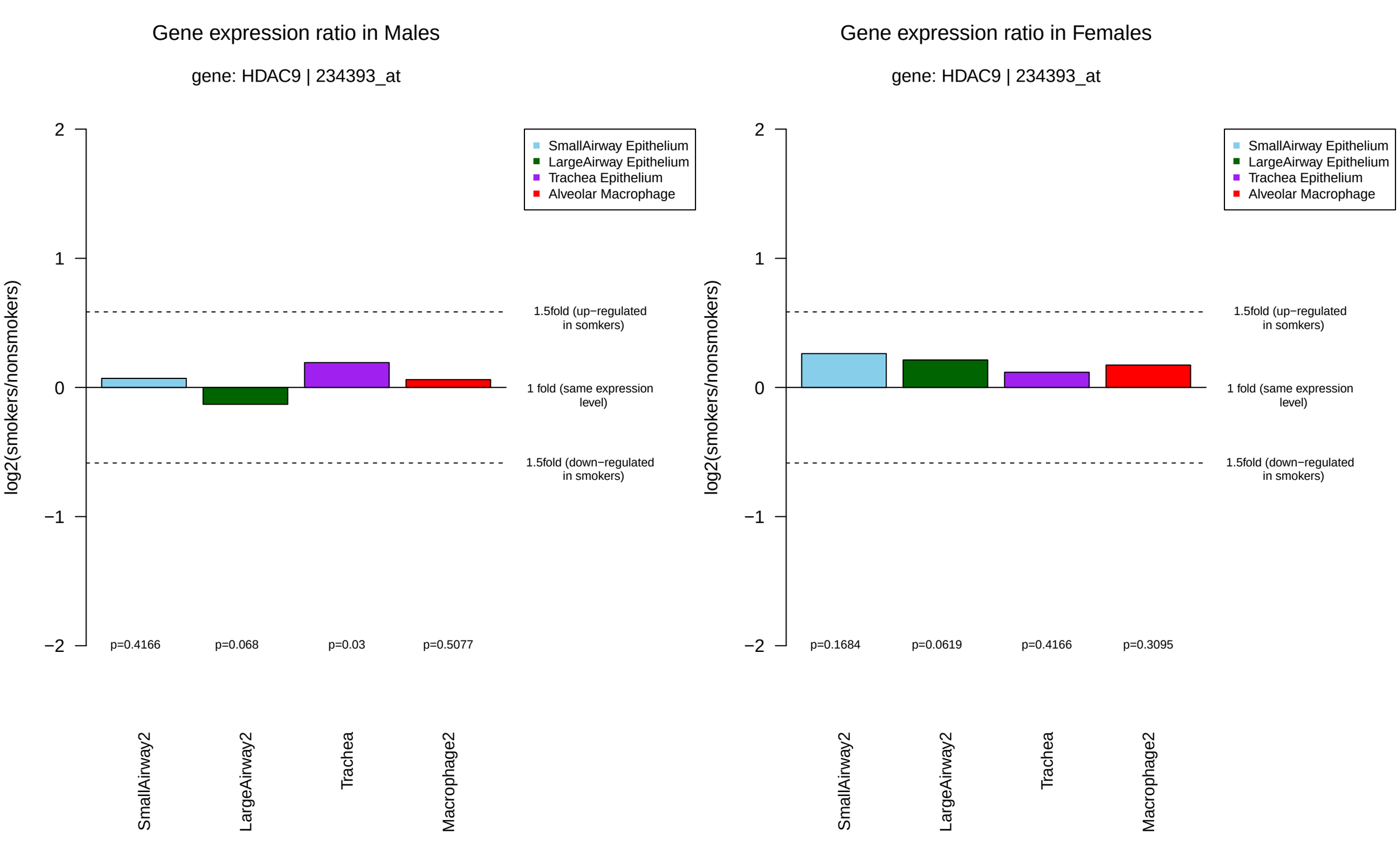
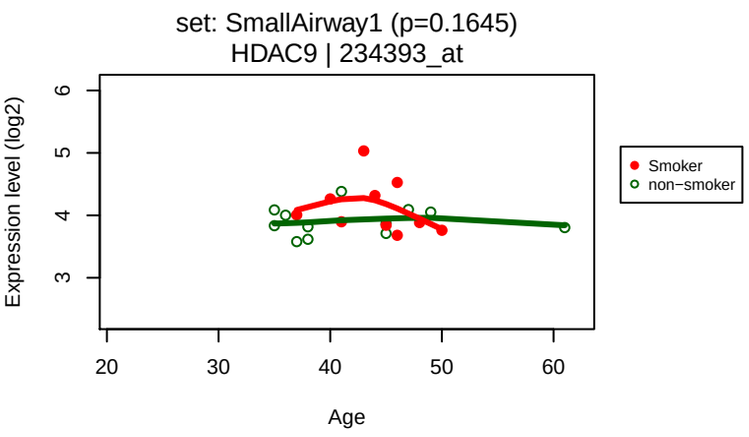
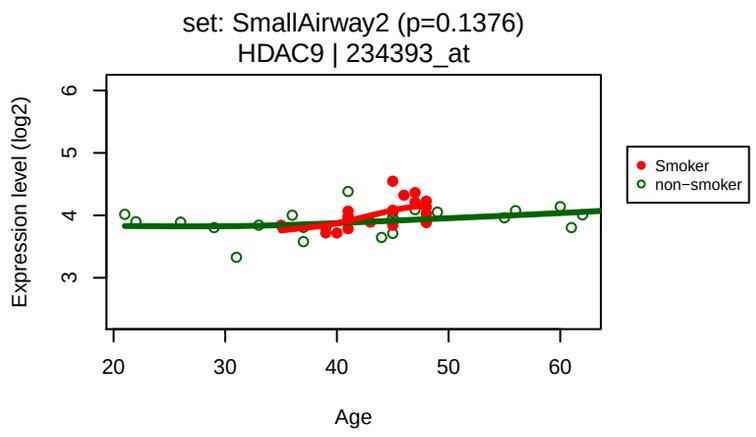
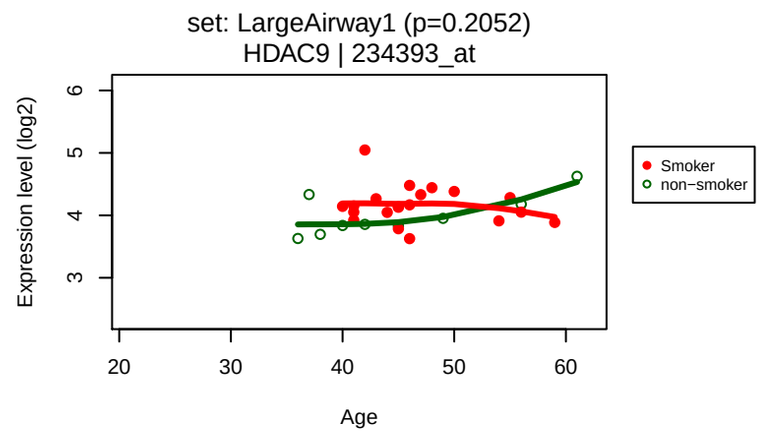

Gene expression of smokers and nonsmokers
gene: HDAC9 | 234393_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1645
p=0.1376
p=0.4344
p=0.2052
p=0.8099
p=0.0143
p=0.9382
p=0.1979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

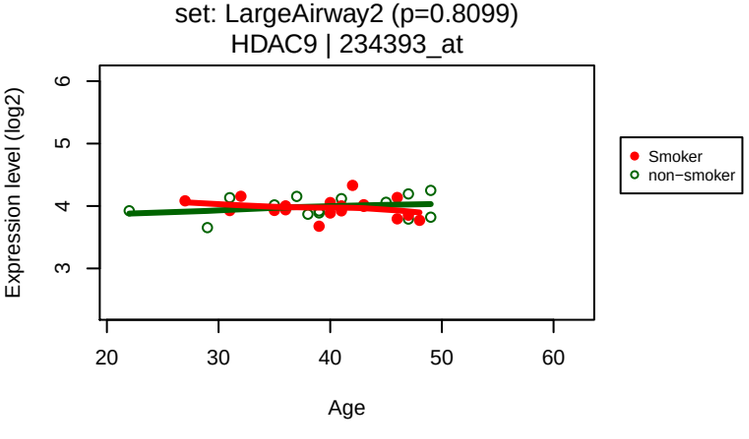

Gene expression ratio (smokers/nonsmokers)
gene: HDAC9 | 234393_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1645
p=0.1376
p=0.4344
p=0.2052
p=0.8099
p=0.0143
p=0.9382
p=0.1979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

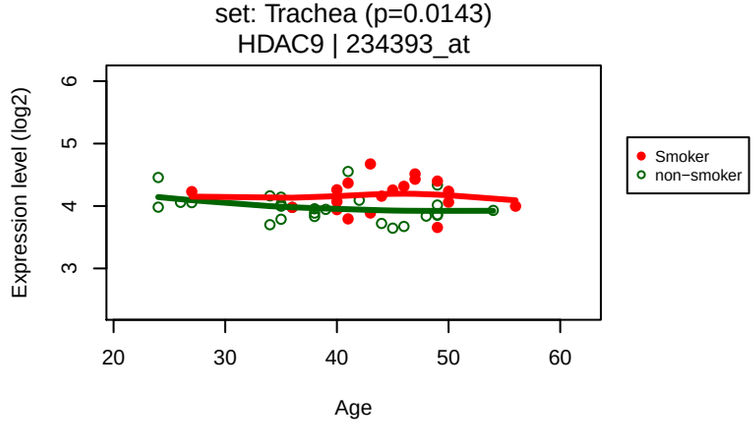

Gene expression ratio in Males
gene: HDAC9 | 234393_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4166
p=0.068
p=0.03
p=0.5077
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HDAC9 | 234393_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1684
p=0.0619
p=0.4166
p=0.3095
SmallAirway2
LargeAirway2
Trachea
Macrophage2

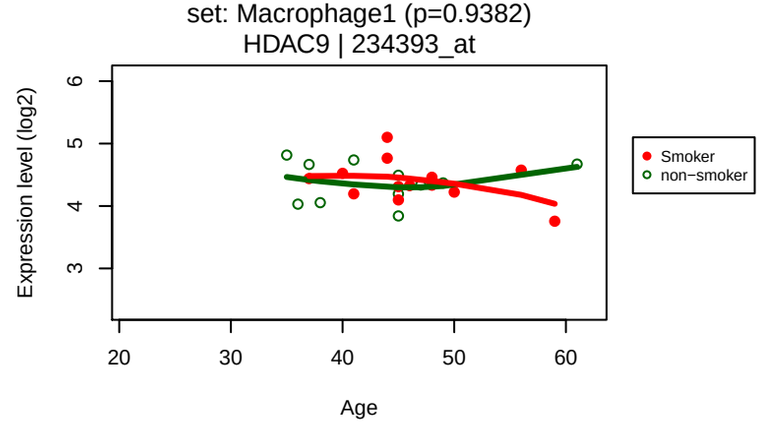

set: SmallAirway1 (p=0.1645)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker

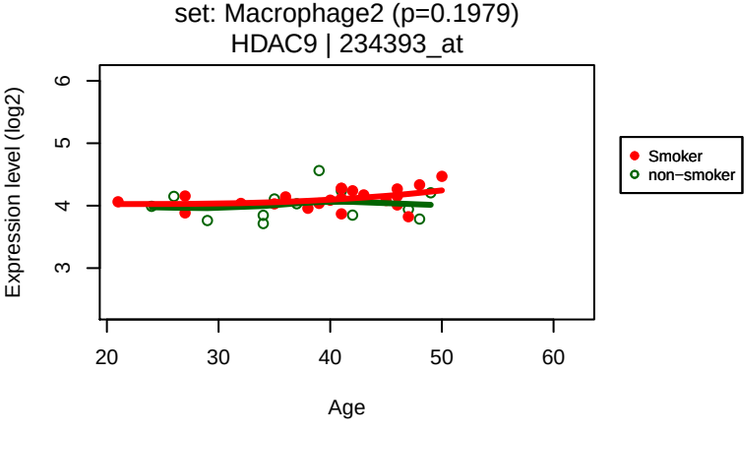

set: SmallAirway2 (p=0.1376)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker

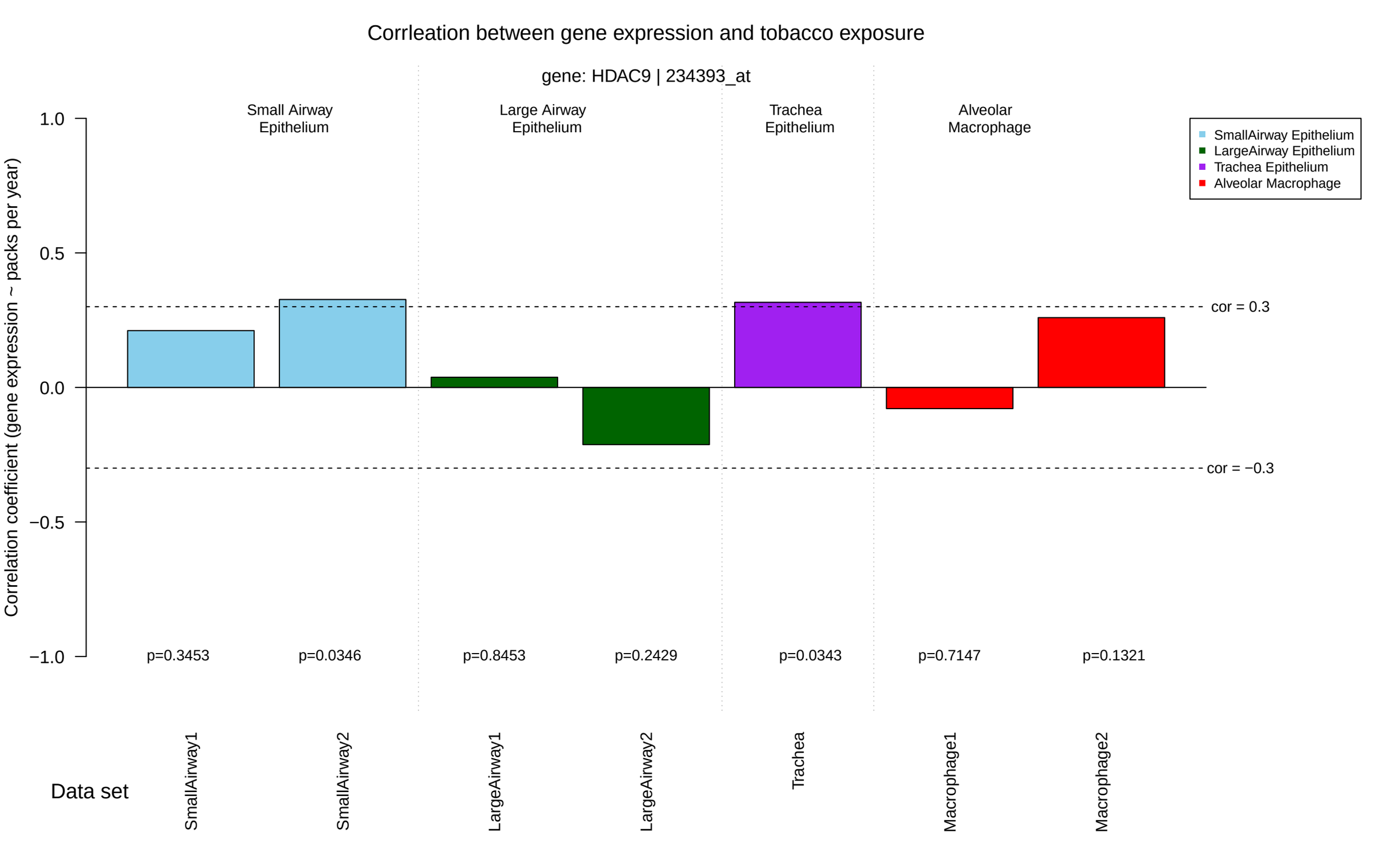

set: LargeAirway1 (p=0.2052)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8099)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0143)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9382)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1979)
HDAC9 | 234393_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HDAC9 | 234393_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3453
p=0.0346
p=0.8453
p=0.2429
p=0.0343
p=0.7147
p=0.1321
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2